Mini Project 1.3: results

1. All tested settings

T = 20 ms, deadline D = 16 ms after each ideal release, 200 jobs per experiment, baseline job body 4 ms of CPU. Release jitter = actual - ideal release. Completion jitter = deviation of (completion - ideal release) from its mean. Times in ms.

Mechanism	Setting	Rel. mean	Rel. std	Rel. max	Comp. std	Comp. max	Hit fraction	Mean R
baseline (control)	no disturbance	0.15	0.03	0.35	0.59	3.03	100.0%	4.61
A: background load	duty 0.25	0.66	0.79	2.57	1.66	4.67	100.0%	4.91
A: background load	duty 0.50	2.28	1.74	5.14	2.08	7.15	100.0%	4.64
A: background load	duty 0.75	2.85	1.99	5.23	2.81	12.63	97.5%	5.22
A: background load	duty 1.00	5.27	0.65	9.15	0.74	4.34	100.0%	4.20
B: delay probability	p=0.02, 5 ms	0.23	0.63	5.39	1.21	8.29	100.0%	4.58
B: delay probability	p=0.05, 5 ms	0.35	1.01	5.35	1.20	6.09	100.0%	4.41
B: delay probability	p=0.10, 5 ms	0.46	1.22	5.41	1.84	8.25	100.0%	4.84
B: delay probability	p=0.20, 5 ms	1.30	2.12	5.42	2.57	7.13	100.0%	4.99
B: delay magnitude	p=0.10, 2 ms	0.29	0.51	2.31	1.05	4.98	100.0%	4.71
B: delay magnitude	p=0.10, 5 ms	0.51	1.32	7.77	1.69	8.48	100.0%	4.71
B: delay magnitude	p=0.10, 8 ms	0.70	1.95	8.41	2.29	10.51	99.5%	4.88
B: delay magnitude	p=0.10, 12 ms	0.95	2.93	12.36	3.17	14.50	93.5%	4.59
baseline (control)	control repeat (sigma=0)	0.14	0.03	0.28	1.07	5.48	100.0%	4.67
C: exec-time variability	lognormal sigma=0.5	0.14	0.03	0.39	2.32	8.83	100.0%	4.85
C: exec-time variability	lognormal sigma=0.8	0.18	0.36	4.72	3.70	19.57	98.5%	4.97
C: exec-time variability	lognormal sigma=1.0	0.39	1.97	24.52	5.91	38.35	94.0%	5.77
C: exec-time variability	lognormal sigma=1.2	1.34	5.61	49.19	10.32	60.93	86.5%	6.91

Mini Project 1.3: results

2. Jitter grew, hit fraction stayed high

Configurations with hit fraction >= 95% and release-jitter std >= 5% of the period (1 ms). These show that meeting deadlines and having low jitter are distinct properties.

Mechanism	Setting	Rel. std (ms)	Rel. max (ms)	Std as % of T	Hit fraction
A: background load	duty 0.50	1.74	5.14	8.7%	100.0%
A: background load	duty 0.75	1.99	5.23	9.9%	97.5%
B: delay probability	p=0.05, 5 ms	1.01	5.35	5.0%	100.0%
B: delay probability	p=0.10, 5 ms	1.22	5.41	6.1%	100.0%
B: delay probability	p=0.20, 5 ms	2.12	5.42	10.6%	100.0%
B: delay magnitude	p=0.10, 5 ms	1.32	7.77	6.6%	100.0%
B: delay magnitude	p=0.10, 8 ms	1.95	8.41	9.8%	99.5%